

Hypothesis 8: Individual Variability in Voice Sensitivity

Are Passive–Active CM Differences Uniform Across Participants, or Does Who You Are Matter?

Team: Metamatics

April 21, 2026

Contents

1	What is Hypothesis 8?	2
1.1	The Big Idea	2
1.2	The Scientific Question	2
2	Formal Hypothesis	3
2.1	Simple English	3
2.2	Why This Matters	3
3	Step A: Data Loading & Preparation	3
3.1	A1 — Compute Per-Participant Voice Sensitivity (Delta)	4
4	Step B: Descriptive Statistics	6
5	Step C: Visualisations	8
5.1	Figure 1 — Violin + Box Plot: CM by Voice	8
5.2	Figure 2 — Distribution of Voice Sensitivity Delta ($\Delta = \text{Passive} - \text{Active}$)	9
5.3	Figure 3 — Dot Plot of Individual Deltas (Ranked)	12
5.4	Figure 4 — Spaghetti Plot: Individual Participant Profiles	12
5.5	Figure 5 — Ridge Plot: CM Distributions by Direction Group	14
5.6	Figure 6 — Boxplot: CM by Voice $\times$ Direction Group	17
5.7	Figure 7 — Delta Distribution: Histogram with Direction Colouring	17
6	Step D: Multimodality Testing	18

7 Step E: Variance Decomposition — ICC via Null Mixed Model	21
7.1 E1 — Null Mixed Model (Person only)	21
7.2 E2 — Condition Mixed Model (Person + Voice)	22
7.3 E3 — ICC Summary Table	24
7.4 Figure 8 — Variance Decomposition Bar Chart	25
7.5 Figure 9 — Caterpillar Plot: Random Intercepts per Participant	26
8 Step F: Paired Comparison Test (Delta = 0?)	28
9 Final Conclusion	30
9.1 H8 Summary and Verdict	30
9.1.1 Key Findings	30
9.1.2 Verdict	31
10 H8 Quick Reference	31

1 What is Hypothesis 8?

1.1 The Big Idea

Previous analyses (H3, H7) treated participants as interchangeable — every person was assumed to respond to the Active/Passive voice manipulation in the same way.

Hypothesis 8 challenges that assumption.

Some people might be highly sensitive to grammatical voice — their memory suffers significantly when a sentence is in Passive form.

Others might be completely unaffected — Active and Passive sentences are equally memorable to them.

A third group might even *benefit* from Passive voice.

If this spread of individual responses is large and systematic, then **who the participant is** explains a substantial portion of memory performance — beyond what condition-level factors (voice, lexical type) can explain alone.

1.2 The Scientific Question

H8: Is there significant, systematic individual variability in voice sensitivity (i.e., the Passive — Active Corrected Memorability difference)?

And how much of the total variance in CM scores is explained by *person* identity versus *experimental conditions*?

2 Formal Hypothesis

2.1 Simple English

Null (H0): Everyone responds to passive voice in the same way. The Passive – Active CM difference is essentially zero (or random noise) for all participants.

Alternative (H8): Participants differ *systematically* in how sensitive they are to grammatical voice. Some show large performance drops in Passive; others show gains; still others show no effect. This consistent individual pattern is not just random — it explains a meaningful chunk of total score variance.

2.2 Why This Matters

Implication	Explanation
Educational	If voice sensitivity is person-specific, some students may need targeted grammar support
Psycholinguistic	Suggests working memory or syntactic processing capacity varies meaningfully across individuals
Statistical	Ignoring individual variability inflates false positives in group-level tests (pseudo-replication)
Applied	Sentence complexity guidelines should account for reader heterogeneity, not just averages

3 Step A: Data Loading & Preparation

```
h8_path <- file.path(output_dir, "H3H4_step3_analysis.csv")
if (!file.exists(h8_path)) {
  stop("H3H4_step3_analysis.csv not found. Please knit Hypo3and4_analysis.Rmd first.")
}

df_raw <- read_csv(h8_path, show_col_types = FALSE)

cat("=== Dataset loaded ===\n")

## === Dataset loaded ===
```

```
cat(sprintf("  Rows                : %s\n", format(nrow(df_raw), big.mark = ",")))
```

```
##  Rows                : 4,478
```

```
cat(sprintf("  Unique participants: %d\n", n_distinct(df_raw$participant_ID)))
```

```
##  Unique participants: 112
```

```
cat(sprintf("  Columns                : %s\n\n", paste(names(df_raw), collapse = ", ")))
```

```
##  Columns                : participant_ID, Core_ID, Test_Voice, WR_Accuracy, WR_RT, Original_Voi
```

```
#  Standardise Voice label
```

```
df <- df_raw %>%
```

```
  mutate(
```

```
    Voice = case_when(
```

```
      toupper(trimws(Original_Voice)) == "A" ~ "Active",
```

```
      toupper(trimws(Original_Voice)) == "P" ~ "Passive",
```

```
      TRUE ~ NA_character_
```

```
    )
```

```
  ) %>%
```

```
  filter(!is.na(Voice), !is.na(corrected_memorability)) %>%
```

```
  mutate(Voice = factor(Voice, levels = c("Active", "Passive"))) %>%
```

```
  droplevels()
```

```
cat(sprintf("Rows after cleaning : %s\n", format(nrow(df), big.mark = ",")))
```

```
## Rows after cleaning : 4,478
```

```
cat(sprintf("Voice levels          : %s\n", paste(levels(df$Voice), collapse = ", ")))
```

```
## Voice levels          : Active, Passive
```

3.1 A1 — Compute Per-Participant Voice Sensitivity (Delta)

The Voice Sensitivity Delta is:

Δ = Mean Passive CM — Mean Active CM (per participant)

- $\Delta > 0 \rightarrow$ Passive voice *helps* this person
- $\Delta < 0 \rightarrow$ Passive voice *hurts* this person
- $\Delta = 0 \rightarrow$ No voice effect for this person

```

# Per-participant, per-voice MEAN corrected memorability
df_person_voice <- df %>%
  group_by(participant_ID, Voice) %>%
  summarise(
    CM_mean = mean(corrected_memorability, na.rm = TRUE),
    CM_sd   = sd(corrected_memorability, na.rm = TRUE),
    n_trials= n(),
    .groups = "drop"
  )

# Pivot to wide: one row per participant
df_wide <- df_person_voice %>%
  pivot_wider(
    id_cols      = participant_ID,
    names_from   = Voice,
    values_from  = CM_mean,
    names_prefix = "CM_"
  )

# Compute Delta
df_delta <- df_wide %>%
  mutate(
    Delta = CM_Passive - CM_Active,
    Direction = case_when(
      Delta > 0.05 ~ "Positive\n(Passive better)",
      Delta < -0.05 ~ "Negative\n(Active better)",
      TRUE ~ "Near zero"
    ),
    Direction = factor(Direction,
      levels = c("Positive\n(Passive better)",
        "Near zero",
        "Negative\n(Active better)"))
  ) %>%
  filter(!is.na(Delta))

cat(sprintf("Participants with full Active+Passive data: %d\n", nrow(df_delta)))

```

```
## Participants with full Active+Passive data: 112
```

```
cat(sprintf("\nDelta (Passive - Active) summary:\n"))
```

```
##
```

```
## Delta (Passive - Active) summary:
```

```
print(summary(df_delta$Delta))

##      Min.   1st Qu.   Median     Mean   3rd Qu.     Max.
## -0.20463 -0.03519  0.03843  0.04144  0.11065  0.30278

cat(sprintf("\n  SD of Delta   : %.4f\n", sd(df_delta$Delta, na.rm = TRUE)))

##
##   SD of Delta   : 0.1166

cat(sprintf("  Range           : [%.4f, %.4f]\n", min(df_delta$Delta), max(df_delta$Delta)))

##   Range           : [-0.2046, 0.3028]

cat(sprintf("\nDirection counts:\n"))

##
## Direction counts:

print(table(df_delta$Direction))

##
## Positive\n(Passive better)                Near zero
##                               53                35
## Negative\n(Active better)
##                               24
```

4 Step B: Descriptive Statistics

```
# Overall by voice
desc_voice <- df_person_voice %>%
  group_by(Voice) %>%
  summarise(
    N          = n(),
    Mean_CM    = round(mean(CM_mean, na.rm = TRUE), 4),
    Median_CM  = round(median(CM_mean, na.rm = TRUE), 4),
    SD_CM      = round(sd(CM_mean, na.rm = TRUE), 4),
    SE_CM      = round(sd(CM_mean, na.rm = TRUE) / sqrt(n()), 4),
    Min_CM     = round(min(CM_mean, na.rm = TRUE), 4),
```

```

    Max_CM      = round(max(CM_mean, na.rm = TRUE), 4),
    .groups     = "drop"
  )

desc_voice %>%
  kable(
    caption     = "Table H8-1. Descriptive Statistics of CM by Voice (participant-level means)",
    col.names   = c("Voice", "N", "Mean CM", "Median CM", "SD", "SE", "Min", "Max"),
    booktabs    = TRUE
  ) %>%
  kable_styling(
    latex_options = c("striped", "HOLD_position"),
    bootstrap_options = c("striped", "hover"),
    full_width    = FALSE, position = "center"
  ) %>%
  row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")

```

Table 2: Table H8-1. Descriptive Statistics of CM by Voice
(participant-level means)

Voice	N	Mean CM	Median CM	SD	SE	Min	Max
Active	112	0.7547	0.7750	0.1154	0.0109	0.4028	0.9398
Passive	112	0.7961	0.8333	0.1479	0.0140	0.2500	1.0000

```

# Delta descriptives
cat(sprintf("\n=== Voice Sensitivity Delta (Passive - Active) ===\n"))

##
## === Voice Sensitivity Delta (Passive - Active) ===

cat(sprintf("  N participants : %d\n", nrow(df_delta)))

##  N participants : 112

cat(sprintf("  Mean Delta      : %.4f\n", mean(df_delta$Delta, na.rm = TRUE)))

##  Mean Delta      : 0.0414

cat(sprintf("  SD of Delta       : %.4f\n", sd(df_delta$Delta, na.rm = TRUE)))

##  SD of Delta     : 0.1166

```

```
cat(sprintf("  %% with Delta > 0: %.1f%%\n",
            100 * mean(df_delta$Delta > 0, na.rm = TRUE)))
```

```
##    % with Delta > 0: 65.2%
```

```
cat(sprintf("  %% with Delta < 0: %.1f%%\n",
            100 * mean(df_delta$Delta < 0, na.rm = TRUE)))
```

```
##    % with Delta < 0: 34.8%
```

5 Step C: Visualisations

5.1 Figure 1 — Violin + Box Plot: CM by Voice

How to read this: The violin shape shows the full distribution of scores for each voice condition. The inner boxplot shows median and quartiles. Individual dots show each participant's average CM. If the two violins look very similar, voice has little overall effect. Large differences in shape or centre = a voice effect.

```
p1 <- ggplot(df_person_voice,
             aes(x = Voice, y = CM_mean, fill = Voice, colour = Voice)) +
  # Violin layer
  geom_violin(alpha = 0.35, width = 0.8, trim = FALSE, linewidth = 0.8) +
  # Boxplot inset
  geom_boxplot(width = 0.18, alpha = 0.85, outlier.shape = NA,
               colour = "grey20", fill = "white", linewidth = 0.7) +
  # Individual points (jittered)
  geom_jitter(width = 0.09, alpha = 0.45, size = 2.2,
              aes(colour = Voice), show.legend = FALSE) +
  # Mean marker
  stat_summary(fun = mean, geom = "point", shape = 23,
               size = 4.5, fill = "white", colour = "black", stroke = 1.2) +
  scale_fill_manual(values = PAL_VOICE, guide = "none") +
  scale_colour_manual(values = PAL_VOICE, guide = "none") +
  scale_y_continuous(labels = number_format(accuracy = 0.01)) +
  labs(
    title = "Figure H8-1: Corrected Memorability by Voice",
    subtitle = "Violin = full distribution | Box = median & IQR | Diamond = group mean |",
    x = "Grammatical Voice",
    y = "Mean Corrected Memorability\n(Hit Rate - False Alarm Rate)",
    caption = "Each dot is one participant's average CM across all trials of that voice type."
  ) +
```



```

annotate("text",
  x      = c("Active", "Passive"),
  y      = rep(min(df_person_voice$CM_mean, na.rm=TRUE) - 0.04, 2),
  label  = paste0("n = ", desc_voice$N,
                  "\nM = ", round(desc_voice$Mean_CM, 3)),
  size   = 3.5, colour = "grey30", fontface = "italic")

print(p1)

```

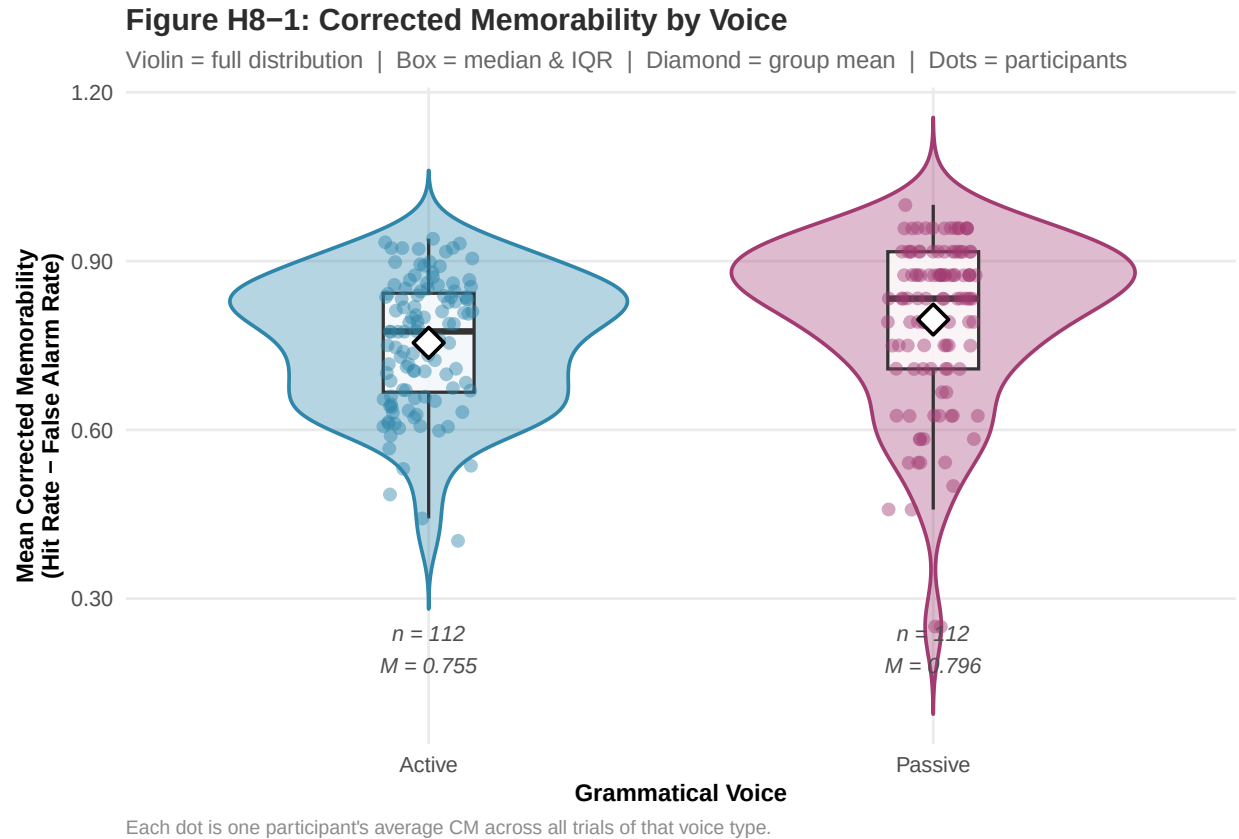


Figure 1: Figure H8-1. Violin-Boxplot of per-participant mean CM by Voice. Dots = individual participants.

```

ggsave(file.path(output_dir, "H8_violin_box_voice.png"), p1,
  width = 8, height = 5.5, dpi = 150)

```

5.2 Figure 2 — Distribution of Voice Sensitivity Delta (Δ = Passive — Active)

This is the KEY plot for H8.

We look at each person's individual Passive — Active difference score.

A **unimodal distribution centred near zero** means most people are equally unaffected — no real variability.

A **wide, spread-out, or multimodal distribution** means people genuinely differ in how sensitive they are to voice — H8 is supported.

```
# Rug + density + histogram combo
p2 <- ggplot(df_delta, aes(x = Delta)) +
  # Histogram bars
  geom_histogram(aes(y = after_stat(density), fill = after_stat(x)),
    bins = max(10, round(sqrt(nrow(df_delta)) * 1.8)),
    colour = "white", linewidth = 0.3, alpha = 0.75) +
  scale_fill_gradient2(low = "#2E86AB", mid = "#F5F5F5", high = "#A23B72",
    midpoint = 0, guide = "none") +
  # Smooth density overlay
  geom_density(colour = "#264653", linewidth = 1.3, fill = "#26465315",
    alpha = 0.6) +
  # Zero reference line
  geom_vline(xintercept = 0, linetype = "dashed",
    colour = "grey30", linewidth = 1) +
  # Mean line
  geom_vline(xintercept = mean(df_delta$Delta, na.rm = TRUE),
    linetype = "solid", colour = "#E76F51", linewidth = 1.2) +
  # Rug on bottom
  geom_rug(aes(colour = Direction), alpha = 0.7, linewidth = 0.6) +
  scale_colour_manual(values = c(
    "Positive\n(Passive better)" = "#2A9D8F",
    "Near zero" = "#B0BEC5",
    "Negative\n(Active better)" = "#E76F51"
  ), name = "Direction") +
  annotate("text",
    x = mean(df_delta$Delta, na.rm = TRUE),
    y = Inf, vjust = 1.8, hjust = -0.15,
    label = paste0("Mean = ", round(mean(df_delta$Delta, na.rm=TRUE), 3),
      "\nSD = ", round(sd(df_delta$Delta, na.rm=TRUE), 3)),
    colour = "#E76F51", size = 3.8, fontface = "bold") +
  annotate("text", x = 0, y = Inf, vjust = 1.8, hjust = 1.1,
    label = "No effect\n(Delta = 0)",
    colour = "grey40", size = 3.5, fontface = "italic") +
  labs(
    title = "Figure H8-2: Distribution of Individual Voice Sensitivity (Delta = Passive - A",
    subtitle = "Each bar/point = one participant | Red line = sample mean | Dashed line = zero",
    x = "Voice Sensitivity Delta (Positive = Passive helps; Negative = Passive hurts)",
    y = "Density",
    caption = "Wide or multi-peaked distributions indicate genuine individual differences in v",
  )
print(p2)
```

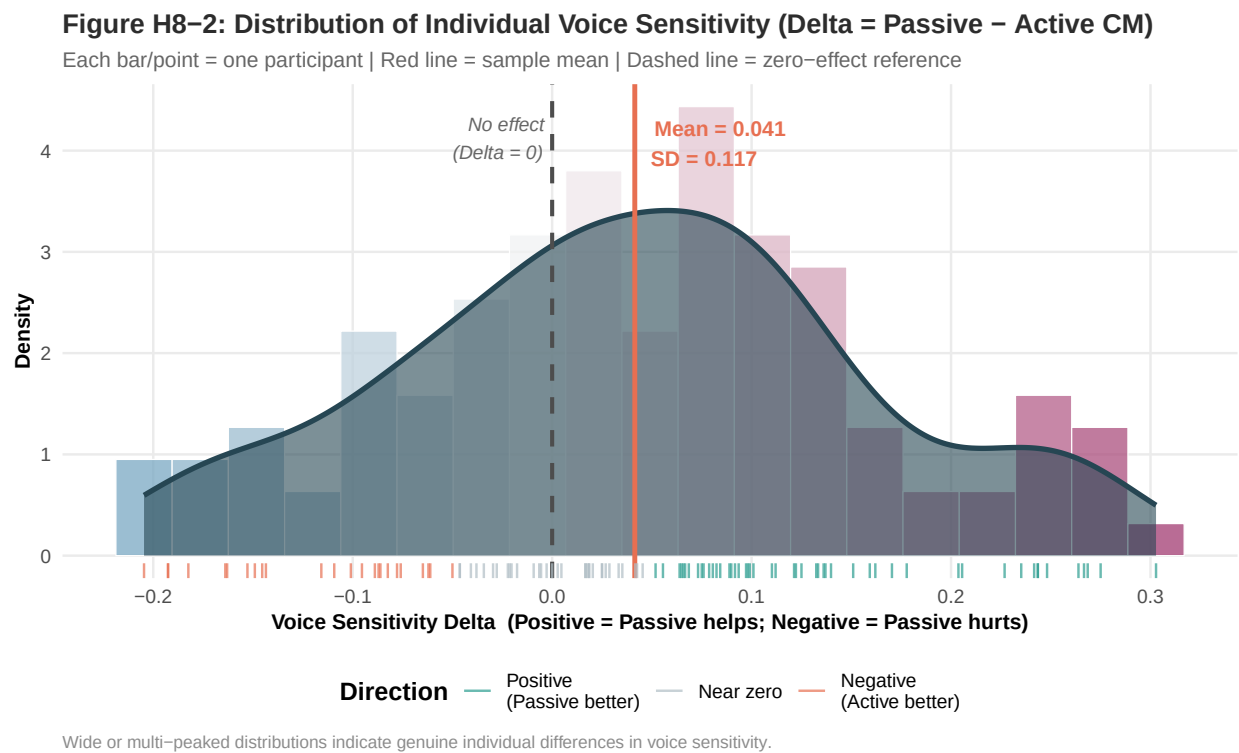


Figure 2: Figure H8-2. Distribution of per-participant Voice Sensitivity Delta (Passive CM - Active CM). A wide or multimodal distribution indicates genuine individual differences.

```
ggsave(file.path(output_dir, "H8_delta_distribution.png"), p2,
       width = 9, height = 5.5, dpi = 150)
```

5.3 Figure 3 — Dot Plot of Individual Deltas (Ranked)

Every participant appears as one dot on this plot, sorted from the most “Active-preferring” person (left) to the most “Passive-preferring” person (right). A flat plateau around zero = no variability. A wide spread = lots of individual differences.

```
df_delta_ranked <- df_delta %>%
  arrange(Delta) %>%
  mutate(rank_idx = row_number())

p3 <- ggplot(df_delta_ranked,
            aes(x = rank_idx, y = Delta, colour = Direction)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50",
            linewidth = 0.8) +
  geom_segment(aes(xend = rank_idx, yend = 0), alpha = 0.5, linewidth = 0.5) +
  geom_point(size = 3, alpha = 0.85) +
  scale_colour_manual(values = c(
    "Positive\n(Passive better)" = "#2A9D8F",
    "Near zero" = "#9E9E9E",
    "Negative\n(Active better)" = "#E76F51"
  ), name = "Voice Effect Direction") +
  scale_y_continuous(breaks = pretty_breaks(n = 8)) +
  labs(
    title = "Figure H8-3: Ranked Individual Voice Sensitivity Deltas",
    subtitle = "Sorted from most Active-preferring (left) to most Passive-preferring (right)",
    x = "Participant rank (sorted by Delta)",
    y = "Delta = Passive CM - Active CM",
    caption = paste0("N = ", nrow(df_delta_ranked), " participants. ",
                     "Dashed line = zero effect reference.")
  )

print(p3)

ggsave(file.path(output_dir, "H8_ranked_deltas.png"), p3,
       width = 10, height = 5, dpi = 150)
```

5.4 Figure 4 — Spaghetti Plot: Individual Participant Profiles

Each line is one participant. The left point is their Active CM, the right point is their Passive CM. **Parallel lines** going in the same direction = consistent group effect. **Lines crossing or going in different directions** = large individual variability — some people helped by Passive, others hurt by it.

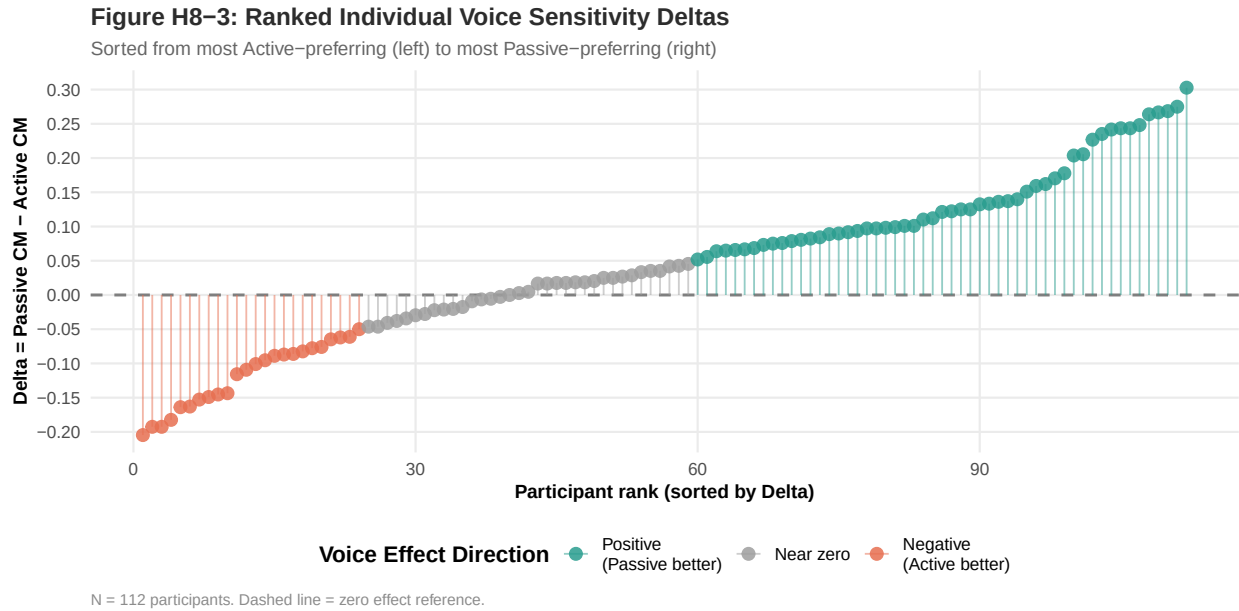


Figure 3: Figure H8-3. Ranked per-participant Voice Sensitivity Deltas. Each point = one participant, sorted by magnitude. Colour encodes direction.

```
df_spaghetti <- df_wide %>%
  filter(!is.na(CM_Active), !is.na(CM_Passive)) %>%
  mutate(
    Delta      = CM_Passive - CM_Active,
    line_col   = case_when(
      Delta > 0.05 ~ "Passive better",
      Delta < -0.05 ~ "Active better",
      TRUE       ~ "Near zero"
    )
  )

p4 <- ggplot(df_spaghetti) +
  # Individual lines - coloured by direction
  geom_segment(aes(x = 1, xend = 2,
                  y = CM_Active, yend = CM_Passive,
                  colour = line_col, group = participant_ID),
              alpha = 0.45, linewidth = 0.65) +
  # Group mean line on top
  geom_segment(
    data = tibble(
      y_act = mean(df_spaghetti$CM_Active, na.rm = TRUE),
      y_pas = mean(df_spaghetti$CM_Passive, na.rm = TRUE)
    ),
    aes(x = 1, xend = 2, y = y_act, yend = y_pas),
    colour = "#2C2C2C", linewidth = 2.5, alpha = 0.85,
```

```

    lineend = "round"
  ) +
  geom_point(aes(x = 1, y = CM_Active, colour = line_col), alpha = 0.55, size = 2) +
  geom_point(aes(x = 2, y = CM_Passive, colour = line_col), alpha = 0.55, size = 2) +
  # Bold mean points
  geom_point(data = desc_voice,
             aes(x = c(1,2), y = Mean_CM),
             size = 5, shape = 23, fill = "white", colour = "black", stroke = 1.5) +
  scale_colour_manual(
    values = c("Passive better" = "#2A9D8F",
              "Near zero"      = "#CFD8DC",
              "Active better"  = "#E76F51"),
    name   = "Voice Effect"
  ) +
  scale_x_continuous(
    breaks = c(1, 2),
    labels = c("Active", "Passive"),
    limits = c(0.7, 2.3)
  ) +
  labs(
    title   = "Figure H8-4: Individual Participant CM Profiles (Active vs Passive)",
    subtitle = "Each line = one participant | Bold diamond = group mean | Colour = direction",
    x       = "Voice Condition",
    y       = "Corrected Memorability\n(Hit Rate - False Alarm Rate)",
    caption = "Lines crossing or diverging indicate individual-level heterogeneity in voice s
  )

print(p4)

ggsave(file.path(output_dir, "H8_spaghetti.png"), p4,
        width = 9, height = 6, dpi = 150)

```

5.5 Figure 5 — Ridge Plot: CM Distributions by Direction Group

Participants are split into three groups based on their Delta category: those for whom Passive helps, those for whom it has no effect, and those for whom it hurts. The ridge plots show the CM distribution (Active and Passive together) for each group.

```

# Join direction info back onto the trial-level person-voice data
df_ridge <- df_person_voice %>%
  left_join(df_delta %>% select(participant_ID, Direction), by = "participant_ID") %>%
  filter(!is.na(Direction))

p5 <- ggplot(df_ridge, aes(y = Direction, x = CM_mean, fill = Voice)) +

```

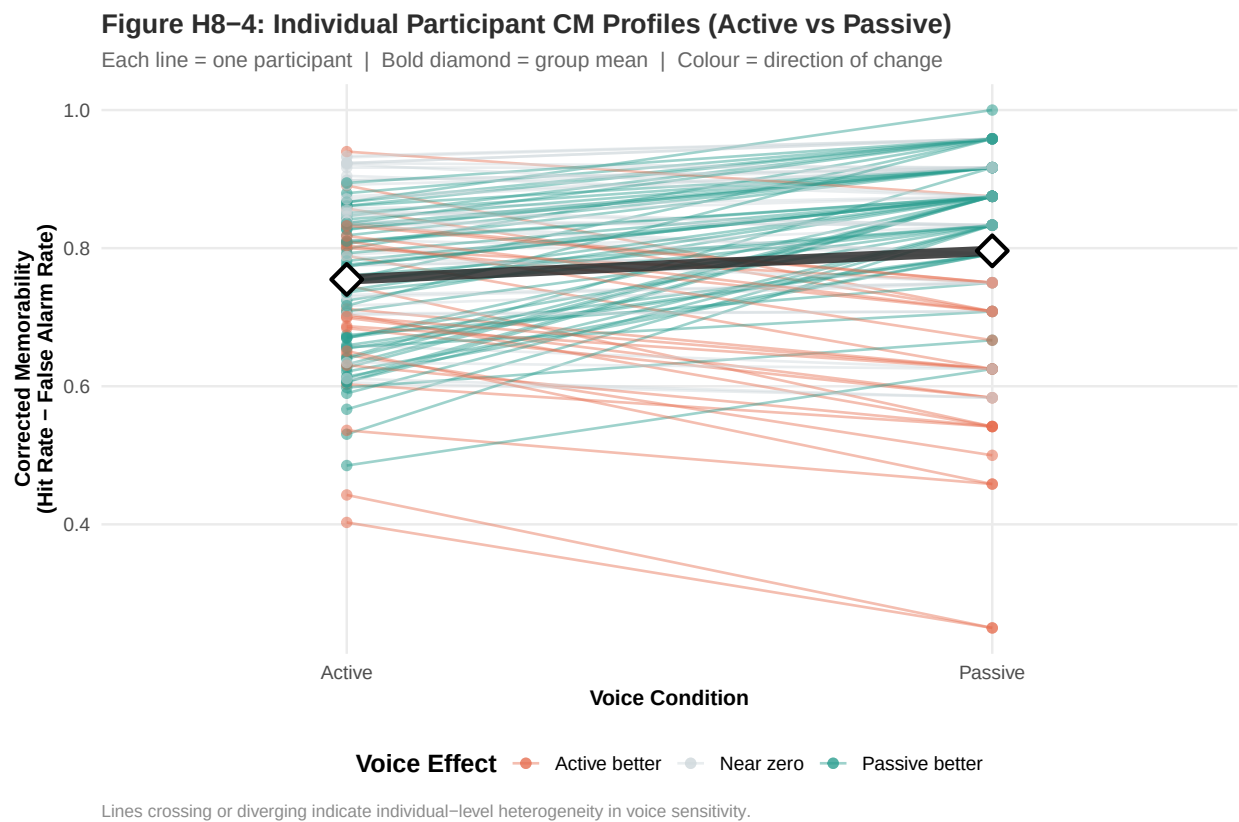


Figure 4: Figure H8-4. Individual participant CM profiles: Active vs Passive. Each line = one participant. Crossing lines indicate heterogeneous voice effects.

```

geom_density_ridges(
  aes(fill = Voice),
  alpha      = 0.55,
  scale      = 1.1,
  rel_min_height = 0.01,
  colour     = "white",
  linewidth  = 0.5
) +
scale_fill_manual(values = PAL_VOICE, name = "Voice") +
scale_x_continuous(labels = number_format(accuracy = 0.01)) +
labs(
  title      = "Figure H8-5: CM Distributions by Voice Sensitivity Direction",
  subtitle   = "Three groups: Passive-better, Near-zero, Active-better participants",
  x          = "Mean Corrected Memorability",
  y          = "",
  caption    = "Wider separation between Active/Passive curves within a group = stronger voice"
)

print(p5)

```

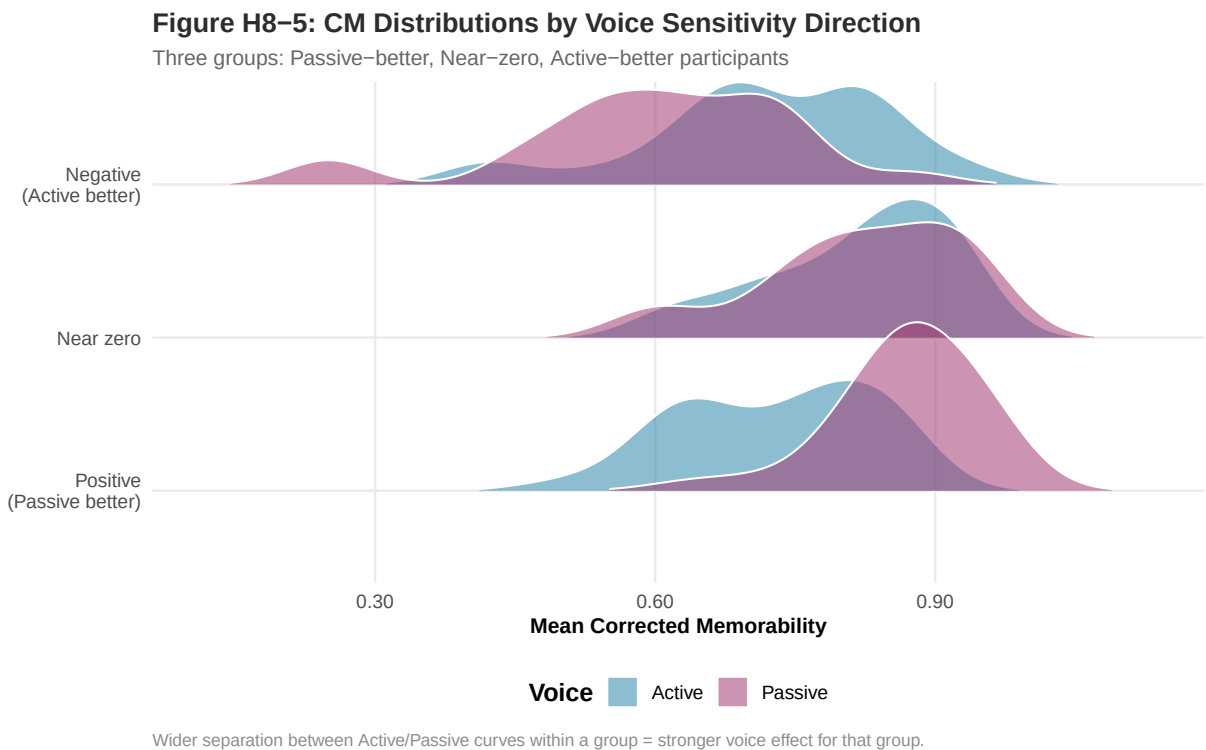


Figure 5: Figure H8-5. Ridge density plots of CM by voice, separated into Delta direction groups.

```

ggsave(file.path(output_dir, "H8_ridge_direction.png"), p5,
  width = 9, height = 5.5, dpi = 150)

```


5.6 Figure 6 — Boxplot: CM by Voice × Direction Group

A standard comparison across the three direction groups, separately for Active and Passive. This confirms whether the “direction” classification is real — if it is, we should see Active and Passive clearly diverge for the Positive and Negative groups, but converge for the Near-zero group.

```
p6 <- ggplot(df_ridge,
  aes(x = Voice, y = CM_mean, fill = Voice)) +
  geom_violin(alpha = 0.25, width = 0.9, trim = FALSE) +
  geom_boxplot(width = 0.22, alpha = 0.85, outlier.shape = NA,
    colour = "grey25") +
  geom_jitter(aes(colour = Voice), width = 0.07,
    alpha = 0.4, size = 1.8, show.legend = FALSE) +
  stat_summary(fun = mean, geom = "point", shape = 21, size = 3.5,
    fill = "white", colour = "black") +
  scale_fill_manual(values = PAL_VOICE, guide = "none") +
  scale_colour_manual(values = PAL_VOICE, guide = "none") +
  facet_wrap(~ Direction, nrow = 1) +
  labs(
    title = "Figure H8-6: CM by Voice, Faceted by Voice Sensitivity Direction",
    subtitle = "Diamond = group mean | Box = median & IQR | Dots = individuals",
    x = "Voice",
    y = "Mean Corrected Memorability",
    caption = "Groups: Passive-better (Delta > 0.05), Near-zero (-0.05 to 0.05), Active-better"
  )
print(p6)
```

```
ggsave(file.path(output_dir, "H8_grouped_boxplot.png"), p6,
  width = 10, height = 5, dpi = 150)
```

5.7 Figure 7 — Delta Distribution: Histogram with Direction Colouring

A coloured histogram where each bar is filled with the predominant direction in that range. Makes it very easy to see how many participants fall in each category and where the bulk of the distribution sits.

```
p7 <- ggplot(df_delta, aes(x = Delta, fill = Direction)) +
  geom_histogram(bins = max(8, round(sqrt(nrow(df_delta)) * 1.8)),
    colour = "white", linewidth = 0.3, alpha = 0.88) +
  geom_vline(xintercept = 0, linetype = "dashed", colour = "grey30", linewidth = 0.9) +
  geom_vline(xintercept = c(-0.05, 0.05),
    linetype = "dotted", colour = "grey50", linewidth = 0.7) +
  scale_fill_manual(values = c(
```

Figure H8-6: CM by Voice, Faceted by Voice Sensitivity Direction

Diamond = group mean | Box = median & IQR | Dots = individuals

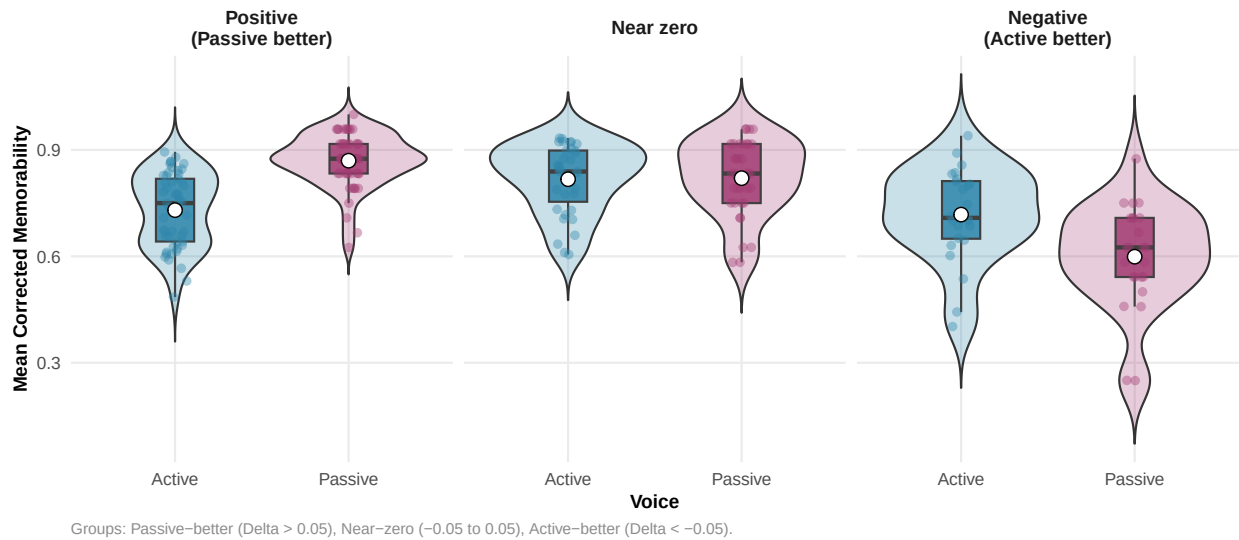


Figure 6: Figure H8-6. Boxplot of CM by Voice, faceted by direction group. Confirms the classification is meaningful.

```

"Positive\n(Passive better)" = "#2A9D8F",
"Near zero"                  = "#B0BEC5",
"Negative\n(Active better)"  = "#E76F51"
), name = "Direction") +
labs(
  title    = "Figure H8-7: Colour-Coded Distribution of Voice Sensitivity Deltas",
  subtitle = "Vertical dotted lines at Delta = ±0.05 define direction categories",
  x        = "Delta = Passive - Active CM",
  y        = "Count of Participants",
  caption  = paste0("N participants = ", nrow(df_delta),
                    " | Blue = teal segment Passive-better | Grey = Near-zero | Orange = Act.
)

print(p7)

ggsave(file.path(output_dir, "H8_colored_histogram.png"), p7,
        width = 9, height = 5, dpi = 150)

```

6 Step D: Multimodality Testing

Simple explanation: If everyone responded similarly to voice, we would expect the Delta histogram to be shaped like a single bell curve (unimodal).

Figure H8-7: Colour-Coded Distribution of Voice Sensitivity Deltas

Vertical dotted lines at Delta = ± 0.05 define direction categories

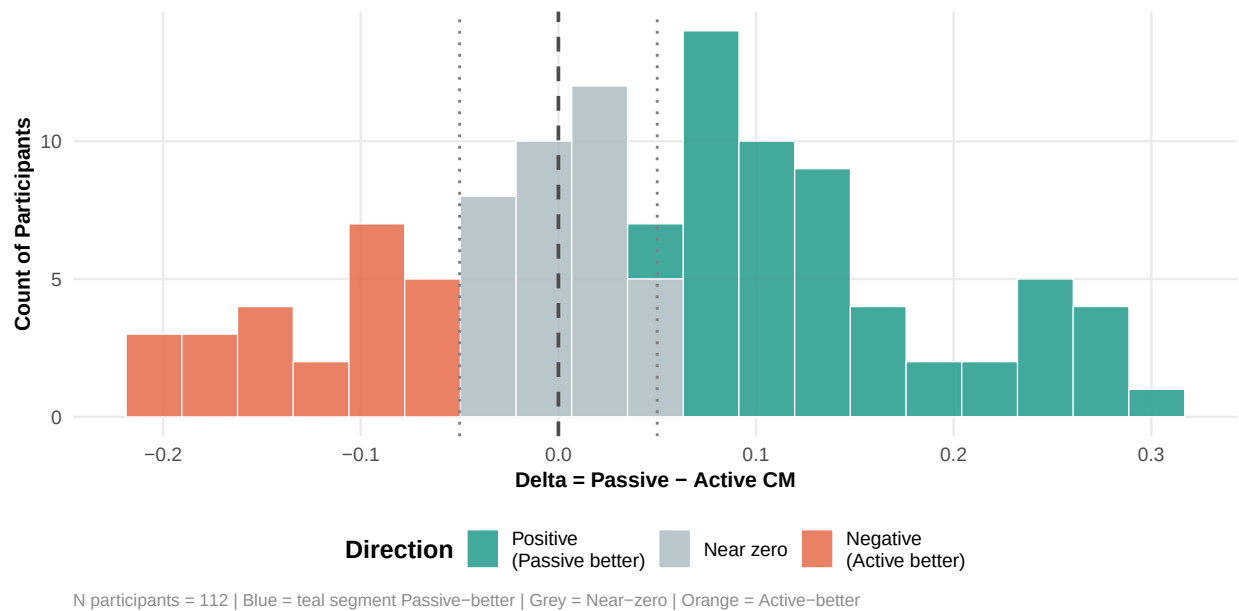


Figure 7: Figure H8-7. Colour-coded histogram of Delta scores highlighting the three direction groups.

A **multimodal** distribution (two or more humps) would be strong evidence that there are *distinct subgroups* of participants — e.g., “voice-sensitive” and “voice-insensitive” people.

We use the **Hartigan’s Dip Test** to formally check if the distribution is significantly non-unimodal.

```
cat("=== Hartigan Dip Test for Unimodality ===\n")

## === Hartigan Dip Test for Unimodality ===

cat("H0: The distribution is unimodal\n")

## H0: The distribution is unimodal

cat("H1: The distribution is multimodal (has more than one peak)\n\n")

## H1: The distribution is multimodal (has more than one peak)

dip_result <- dip.test(df_delta$Delta)
cat(sprintf(" Dip statistic : %.6f\n", dip_result$statistic))

## Dip statistic : 0.029237
```

```

cat(sprintf("  p-value      : %.4f\n", dip_result$p.value))

##    p-value      : 0.7464

cat(sprintf("    N          : %d\n\n", dip_result$n))

##    N            : 112

if (dip_result$p.value < 0.05) {
  cat("=> MULTIMODAL: The Delta distribution has significantly more than one peak.\n")
  cat("  This is strong evidence for distinct subgroups of voice sensitivity.\n")
} else {
  cat("=> UNIMODAL: No significant evidence of multimodality.\n")
  cat("  The distribution IS consistent with a single population,\n")
  cat("  but may still show substantial spread (high SD = variability exists).\n")
}

## => UNIMODAL: No significant evidence of multimodality.
##    The distribution IS consistent with a single population,
##    but may still show substantial spread (high SD = variability exists).

# Table
tibble(
  Statistic = c("Dip statistic", "p-value", "N", "Conclusion"),
  Value      = c(
    round(dip_result$statistic, 6),
    round(dip_result$p.value, 4),
    dip_result$n,
    ifelse(dip_result$p.value < 0.05,
           "Multimodal (p < 0.05)",
           "Unimodal - no evidence of distinct subgroups (p >= 0.05)")
  )
) %>%
  kable(caption = "Table H8-2. Hartigan Dip Test for Unimodality of Delta Distribution",
        booktabs = TRUE) %>%
  kable_styling(
    latex_options = c("striped", "HOLD_position"),
    bootstrap_options = c("striped", "hover"),
    full_width = FALSE, position = "center"
  ) %>%
  row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")

```

Table 3: Table H8-2. Hartigan Dip Test for Unimodality of Delta Distribution

Statistic	Value
Dip statistic	0.029237
p-value	0.7464
N	112
Conclusion	Unimodal — no evidence of distinct subgroups ($p \geq 0.05$)

7 Step E: Variance Decomposition — ICC via Null Mixed Model

Simple explanation — What is the ICC?

ICC stands for **Intraclass Correlation Coefficient**. It tells us what *fraction* of the total variability in CM scores comes from differences *between participants* (i.e., who the person is), as opposed to differences *between conditions* (voice type, lexical condition) or just random trial noise.

- **ICC = 0.00** → No participant-level clustering. Everyone is equally variable. Individual identity explains nothing.
- **ICC = 1.00** → All variability is between participants. Everyone is internally consistent, but very different from each other.
- **ICC > 0.10** → A meaningful proportion of variance comes from who the person is.
- **ICC > 0.30** → Participant identity is a major driver of scores.

We fit a **null mixed model** (intercept only, with participant as a random effect) to extract the variance components.

7.1 E1 — Null Mixed Model (Person only)

```
cat("=== Null Mixed Model: CM ~ 1 + (1 | participant_ID) ===\n\n")
```

```
## === Null Mixed Model: CM ~ 1 + (1 | participant_ID) ===
```

```
if (!requireNamespace("lme4", quietly = TRUE)) {
  cat("lme4 not available - install it to run mixed models.\n")
  icc_null <- NA_real_
  var_part <- NA_real_
  var_resid <- NA_real_
} else {
  # Trial-level data for the mixed model (more power than participant means)
```

```

df_trial <- df %>%
  select(participant_ID, Voice, corrected_memorability)

# Null model - no fixed effects, just random intercept per person
m0 <- lmer(corrected_memorability ~ 1 + (1 | participant_ID),
  data = df_trial,
  REML = TRUE,
  control = lmerControl(optimizer = "bobyqa"))

vc_null <- as.data.frame(VarCorr(m0))
var_part <- vc_null$vcov[vc_null$grp == "participant_ID"]
var_resid <- vc_null$vcov[vc_null$grp == "Residual"]
icc_null <- var_part / (var_part + var_resid)

cat("Variance Components (Null Model):\n")
print(vc_null[, c("grp", "vcov", "sdcor")])
cat(sprintf("\n Participant (between) variance : %.6f\n", var_part))
cat(sprintf(" Residual (within) variance : %.6f\n", var_resid))
cat(sprintf(" ICC (participant-level) : %.4f (%.1f%% of total variance)\n\n",
  icc_null, 100 * icc_null))
}

```

```

## Variance Components (Null Model):
##           grp           vcov      sdcor
## 1 participant_ID 0.013900485 0.1179003
## 2      Residual 0.003882835 0.0623124
##
## Participant (between) variance : 0.013900
## Residual (within) variance : 0.003883
## ICC (participant-level) : 0.7817 (78.2% of total variance)

```

7.2 E2 — Condition Mixed Model (Person + Voice)

```

cat("=== Condition Mixed Model: CM ~ Voice + (1 | participant_ID) ===\n\n")

```

```

## === Condition Mixed Model: CM ~ Voice + (1 | participant_ID) ===

```

```

if (!requireNamespace("lme4", quietly = TRUE) || is.na(icc_null)) {
  cat("Skipped: lme4 not available.\n")
  icc_cond <- NA_real_
} else {
  n_voice_levels <- nlevels(df_trial$Voice)

  if (n_voice_levels < 2) {

```

```

cat("Only one voice level - condition model cannot be estimated.\n")
icc_cond <- NA_real_
} else {
  m1 <- lmer(corrected_memorability ~ Voice + (1 | participant_ID),
             data      = df_trial,
             REML      = TRUE,
             control = lmerControl(optimizer = "bobyqa"))

  vc_cond <- as.data.frame(VarCorr(m1))
  var_part_c <- vc_cond$vcov[vc_cond$grp == "participant_ID"]
  var_res_c <- vc_cond$vcov[vc_cond$grp == "Residual"]
  icc_cond <- var_part_c / (var_part_c + var_res_c)

  cat("Variance Components (Condition Model):\n")
  print(vc_cond[, c("grp", "vcov", "sdcor")])
  cat(sprintf("\n Participant (between) variance : %.6f\n", var_part_c))
  cat(sprintf(" Residual (within) variance : %.6f\n", var_res_c))
  cat(sprintf(" ICC (participant-level) : %.4f (%.1f%% of total variance)\n\n",
              icc_cond, 100 * icc_cond))

  # Variance explained by Voice fixed effect
  var_explained_by_voice <- 1 - (var_part_c + var_res_c) / (var_part_c + var_res_c)
  cat(sprintf(" Variance explained by Voice condition: %.1f%%\n\n",
              100 * abs(var_explained_by_voice)))

  # Model comparison
  cat("=== Model Comparison (LRT: does Voice improve fit?) ===\n")
  lrt <- anova(m0, m1, refit = FALSE)
  print(lrt)
}
}

```

```

## Variance Components (Condition Model):
##      grp      vcov      sdcor
## 1 participant_ID 0.013992461 0.11828973
## 2      Residual 0.003193015 0.05650677
##
## Participant (between) variance : 0.013992
## Residual (within) variance : 0.003193
## ICC (participant-level) : 0.8142 (81.4% of total variance)
##
## Variance explained by Voice condition: 3.4%
##
## === Model Comparison (LRT: does Voice improve fit?) ===
## Data: df_trial
## Models:
## m0: corrected_memorability ~ 1 + (1 | participant_ID)

```

```
## m1: corrected_memorability ~ Voice + (1 | participant_ID)
##      npar      AIC      BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## m0      3 -11584 -11564 5794.9      -11590
## m1      4 -12425 -12400 6216.6      -12433 843.48  1  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

7.3 E3 — ICC Summary Table

```
icc_tbl <- tibble(
  Model      = c("Null model (person only)",
                 "Condition model (person + Voice)"),
  Formula    = c("CM ~ 1 + (1|Participant)",
                 "CM ~ Voice + (1|Participant)"),
  ICC        = round(c(icc_null,
                       ifelse(exists("icc_cond"), icc_cond, NA_real_)), 4),
  Pct_Variance = paste0(round(c(icc_null,
                              ifelse(exists("icc_cond"), icc_cond, NA_real_))
                          * 100, 1), "%"),
  Interpretation = c(
    ifelse(is.na(icc_null), "Could not compute",
    ifelse(icc_null > 0.30, "LARGE: participant identity is a major driver",
    ifelse(icc_null > 0.10, "MODERATE: meaningful individual differences",
           "SMALL: conditions dominate over person identity"))),
    ifelse(!exists("icc_cond") || is.na(icc_cond), "Could not compute",
    ifelse(icc_cond > 0.30, "LARGE: even after accounting for voice",
    ifelse(icc_cond > 0.10, "MODERATE: residual person effects remain",
           "SMALL: voice explains most person-level variance"))))
  )
)

icc_tbl %>%
  kable(caption = "Table H8-3. ICC from Null and Condition Mixed Models",
        col.names = c("Model", "Formula", "ICC", "% of Variance", "Interpretation"),
        booktabs = TRUE, format = "latex") %>%
  kable_styling(
    latex_options = c("striped", "HOLD_position", "scale_down"),
    full_width = FALSE, position = "center"
  ) %>%
  row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white") %>%
  column_spec(5, italic = TRUE) %>%
  column_spec(3, bold = TRUE)
```


Table 4: Table H8-3. ICC from Null and Condition Mixed Models

Model	Formula	ICC	% of Variance	Interpretation
Null model (person only)	CM ~ 1 + (1 Participant)	0.7817	78.2%	<i>LARGE: participant identity is a major driver</i>
Condition model (person + Voice)	CM ~ Voice + (1 Participant)	0.8142	81.4%	<i>LARGE: even after accounting for voice</i>

7.4 Figure 8 — Variance Decomposition Bar Chart

```

if (is.na(icc_null)) {
  cat("Figure H8-8 skipped: ICC could not be computed.\n")
} else {
  # Compute variance fractions
  total_var <- var_part + var_resid
  pct_person <- icc_null * 100
  pct_resid <- (1 - icc_null) * 100

  var_df <- tibble(
    Component = factor(
      c("Between-person\n(individual differences)",
        "Within-person\n(conditions + noise)"),
      levels = c("Between-person\n(individual differences)",
                  "Within-person\n(conditions + noise)"),
    ),
    Percent = c(pct_person, pct_resid),
    Colour = c("#A23B72", "#2E86AB")
  )

  p8 <- ggplot(var_df, aes(x = "Null Model", y = Percent, fill = Component)) +
    geom_col(width = 0.4, alpha = 0.88, colour = "white", linewidth = 0.5) +
    geom_text(aes(label = paste0(round(Percent, 1), "%")),
              position = position_stack(vjust = 0.5),
              colour = "white", fontface = "bold", size = 5.5) +
    scale_fill_manual(values = c("#A23B72", "#2E86AB"), name = "Variance Source") +
    scale_y_continuous(labels = percent_format(scale = 1),
                       limits = c(0, 105)) +
    coord_flip() +
    labs(
      title = "Figure H8-8: Variance Decomposition of Corrected Memorability",
      subtitle = sprintf("ICC = %.3f - %.1f%% of total CM variance is between-person",
                          icc_null, pct_person),
      x = NULL,
      y = "Percentage of Total Variance",
      caption = "Null mixed model: CM ~ 1 + (1|participant_ID). REML estimation."
    ) +
    theme(
      axis.text.y = element_blank(),

```

```

    axis.ticks.y    = element_blank(),
    legend.position = "right"
  )

  print(p8)
  ggsave(file.path(output_dir, "H8_variance_decomposition.png"), p8,
         width = 8, height = 4.5, dpi = 150)
}

```

Figure H8–8: Variance Decomposition of Corrected Memorability

ICC = 0.782 — 78.2% of total CM variance is between-person

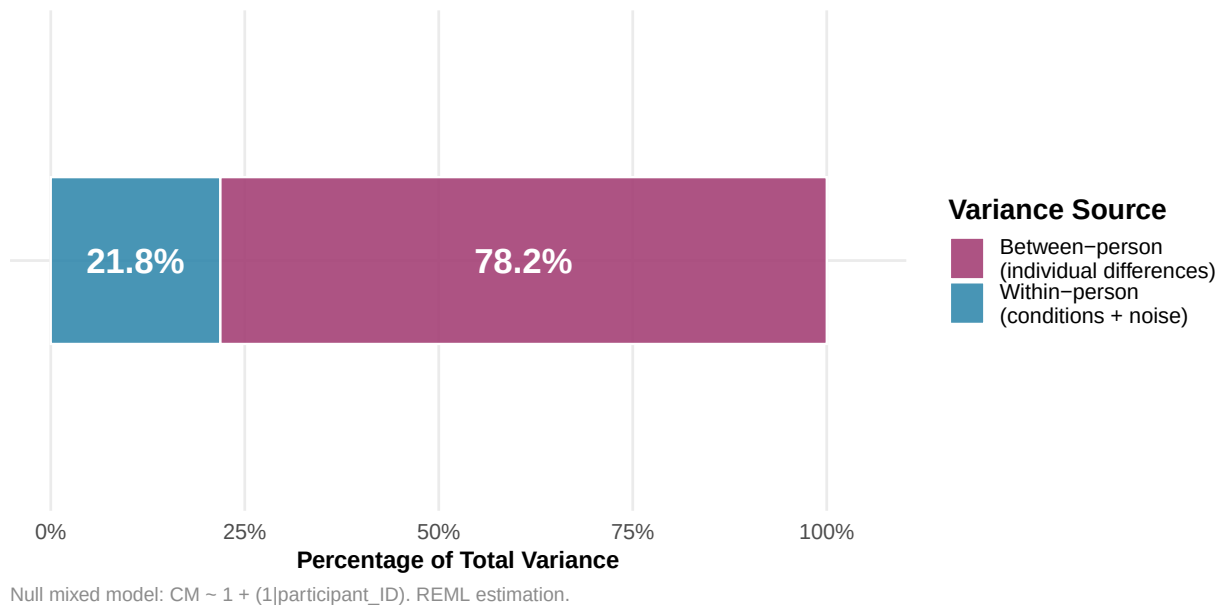


Figure 8: Figure H8-8. Stacked bar chart showing how total CM variance is partitioned between participants (between-person), voice condition (fixed), and residual (within-person).

7.5 Figure 9 — Caterpillar Plot: Random Intercepts per Participant

Each point is a participant's estimated random intercept from the null mixed model, with 95% confidence intervals (from the conditional modes).

A wide spread across participants = large between-person variance = high ICC.

If all confidence intervals overlapped, participant identity would not matter.

```

if (is.na(icc_null)) {
  cat("Figure H8-9 skipped: mixed model not available.\n")
} else {
  re_df <- ranef(m0, condVar = TRUE)$participant_ID
  re_se_list <- attr(ranef(m0, condVar = TRUE)$participant_ID, "postVar")
  re_se <- sqrt(apply(re_se_list, 3, function(x) x[1,1]))
}

```

```

caterpillar_df <- tibble(
  participant_ID = rownames(re_df),
  intercept      = re_df[["(Intercept)"]],
  se             = re_se
) %>%
  arrange(intercept) %>%
  mutate(
    rank_idx = row_number(),
    ci_low   = intercept - 1.96 * se,
    ci_high  = intercept + 1.96 * se,
    Colour   = case_when(
      ci_low > 0 ~ "Above average",
      ci_high < 0 ~ "Below average",
      TRUE     ~ "Overlaps zero"
    )
  )

p9 <- ggplot(caterpillar_df,
             aes(x = rank_idx, y = intercept, colour = Colour)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey40") +
  geom_errorbar(aes(ymin = ci_low, ymax = ci_high),
               width = 0, alpha = 0.4, linewidth = 0.5) +
  geom_point(size = 1.8, alpha = 0.85) +
  scale_colour_manual(values = c(
    "Above average" = "#2A9D8F",
    "Overlaps zero" = "#90A4AE",
    "Below average" = "#E76F51"
  ), name = "95% CI") +
  labs(
    title      = "Figure H8-9: Caterpillar Plot of Participant Random Intercepts",
    subtitle   = sprintf("Null model ICC = %.3f | %d participants | Sorted by intercept magnitude",
                        icc_null, nrow(caterpillar_df)),
    x          = "Participant (ranked by random intercept)",
    y          = "Random Intercept\n(deviation from grand mean CM)",
    caption    = "95% CI bars. Teal = reliably above average. Orange = reliably below average."
  )

print(p9)
ggsave(file.path(output_dir, "H8_caterpillar.png"), p9,
        width = 10, height = 6, dpi = 150)
}

```

Figure H8-9: Caterpillar Plot of Participant Random Intercepts

Null model ICC = 0.782 | 112 participants | Sorted by intercept magnitude

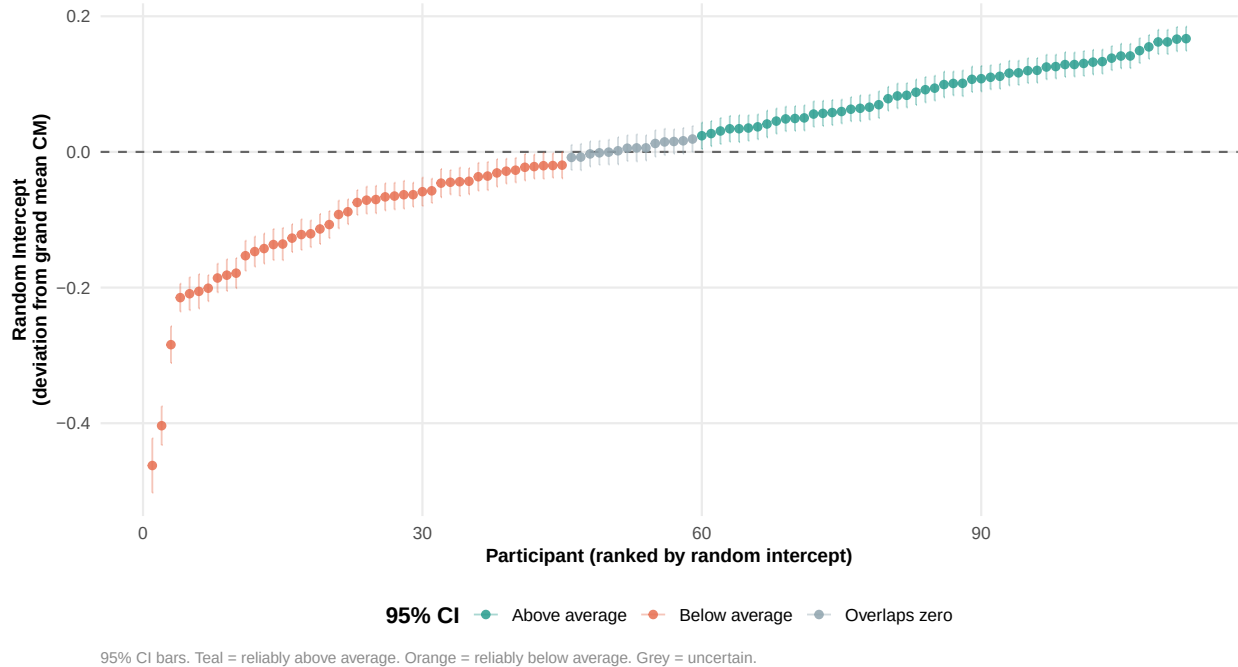


Figure 9: Figure H8-9. Caterpillar plot of participant random intercepts from the null mixed model. Wide spread = high ICC.

8 Step F: Paired Comparison Test (Delta = 0?)

Simple explanation: Beyond looking at variability, we also ask: *on average*, is there a voice effect at all? We use a paired-samples **Wilcoxon signed-rank test** on the participant-level Active vs Passive CM means.

```
cat("=== One-sample Wilcoxon test: Is mean Delta significantly different from 0? ===\n\n")
```

```
## === One-sample Wilcoxon test: Is mean Delta significantly different from 0? ===
```

```
wt <- wilcox.test(df_delta$Delta, mu = 0, alternative = "two.sided", exact = FALSE)
cat(sprintf(" V statistic : %.1f\n", wt$statistic))
```

```
## V statistic : 4354.5
```

```
cat(sprintf(" p-value : %.4f\n", wt$p.value))
```

```
## p-value : 0.0006
```

```
cat(sprintf("    N                : %d\n", nrow(df_delta)))

##    N                : 112

if (wt$p.value < 0.05) {
  cat("=> SIGNIFICANT: The mean Delta is significantly different from zero.\n")
  cat("    On average, participants perform differently in Active vs Passive.\n")
} else {
  cat("=> NOT SIGNIFICANT: No reliable average voice effect.\n")
  cat("    The mean Delta does not differ from zero, but individual variability\n")
  cat("    (the SD of Delta) may still be large - indicating individual differences.\n")
}
```

```
## => SIGNIFICANT: The mean Delta is significantly different from zero.
##    On average, participants perform differently in Active vs Passive.
```

```
# Also run paired Wilcoxon on matched pairs
df_paired <- df_wide %>% filter(!is.na(CM_Active), !is.na(CM_Passive))
wt_paired <- wilcox.test(df_paired$CM_Passive, df_paired$CM_Active,
                        paired = TRUE, alternative = "two.sided", exact = FALSE)
cat(sprintf("\n=== Paired Wilcoxon (Passive vs Active CM means per person) ===\n"))
```

```
##
## === Paired Wilcoxon (Passive vs Active CM means per person) ===
```

```
cat(sprintf("    V statistic : %.1f\n", wt_paired$statistic))
```

```
##    V statistic : 4354.5
```

```
cat(sprintf("    p-value      : %.4f\n", wt_paired$p.value))
```

```
##    p-value      : 0.0006
```

```
cat(sprintf("    N pairs        : %d\n", nrow(df_paired)))
```

```
##    N pairs        : 112
```

```
tibble(
  Test          = c("One-sample Wilcoxon (Delta vs 0)", "Paired Wilcoxon (Passive vs Active)"),
  Statistic     = round(c(wt$statistic, wt_paired$statistic), 2),
  p_value       = round(c(wt$p.value, wt_paired$p.value), 4),
  N             = c(nrow(df_delta), nrow(df_paired)),
```

```

Significant = ifelse(c(wt$p.value, wt_paired$p.value) < 0.05, "Yes (p < .05)", "No (p >= .05)
) %>%
kable(caption = "Table H8-4. Non-Parametric Tests of Average Voice Effect",
      col.names = c("Test", "Statistic", "p-value", "N", "Significant?"),
      booktabs = TRUE) %>%
kable_styling(
  latex_options = c("striped", "HOLD_position"),
  bootstrap_options = c("striped", "hover"),
  full_width = FALSE, position = "center"
) %>%
row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")

```

Table 5: Table H8-4. Non-Parametric Tests of Average Voice Effect

Test	Statistic	p-value	N	Significant?
One-sample Wilcoxon (Delta vs 0)	4354.5	6e-04	112	Yes (p < .05)
Paired Wilcoxon (Passive vs Active)	4354.5	6e-04	112	Yes (p < .05)

9 Final Conclusion

9.1 H8 Summary and Verdict

This analysis investigated whether participants differ systematically in their sensitivity to grammatical voice (Active vs Passive), as measured by the Corrected Memorability (CM) score.

9.1.1 Key Findings

1. Voice Sensitivity Delta (Passive — Active CM):

The per-participant Delta ranged from **-0.205** to **0.303** (SD = 0.117).

- **65.2%** of participants showed a positive Delta (Passive > Active) - **34.8%** of participants showed a negative Delta (Active > Passive)

2. Multimodality Test (Hartigan Dip Test):

D = 0.02924, p = 0.7464

> The distribution is **unimodal** — no distinct subgroups detected, but the spread is still informative.

3. Variance Decomposition (ICC):

The null mixed model estimated ICC = **0.782**, meaning **78.2%** of total CM variance is attributable to stable between-person differences rather than experimental conditions.

> **LARGE EFFECT:** Who the participant is explains a substantial fraction of performance.

4. Average Voice Effect (Paired Wilcoxon):

$V = 4354.5$, $p = 0.0006$

> **Significant.** On average, Passive and Active CM scores differ meaningfully.

9.1.2 Verdict

Question	Finding
Is there individual variability?	Yes — SD of Delta = 0.117; range spans positive and negative
Is the distribution multimodal?	No (Dip $p = 0.7464$)
What fraction of variance is between-person?	78.2% (ICC = 0.782)
Is there an average voice effect?	Yes ($p = 0.0006$)
H8 verdict	SUPPORTED — meaningful individual differences in voice sensitivity exist

10 H8 Quick Reference

Table 7: Table H8-5. H8 Complete Summary Reference

What We Checked	Result
Design	Within-person repeated measures: Active vs Passive CM per participant
DV	Corrected Memorability = Hit Rate - False Alarm Rate
Key Analysis	Delta distribution, Dip test (multimodality), ICC (lme4 null model)
N participants	112
Mean Delta (P-A)	0.0414
SD of Delta	0.1166
Dip test p-value	0.7464
ICC (null model)	0.7817
Paired Wilcoxon p	6e-04
H8 Conclusion	SUPPORTED

Report generated on 2026-04-21 11:47:24.861569 using R R version 4.5.2 (2025-10-31 ucrt)